

TRIAPERFORMANCE GUIDE

The Fuel Kit

How to fuel your training and your racing

It covers one thing, and it covers it completely: how to fuel your sessions and your races.

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What this guide covers

This is training material. It is not medical advice, and it is not individual nutrition advice.

It covers one thing, and it covers it completely: **how to fuel your sessions and your races.**

It does not cover diets, weight loss, body composition or supplementation. Not because those don't matter, but because they aren't my job. I'm your coach, not your dietitian, and that distinction isn't a technicality — it's the line between what I can defend from experience and what requires a licence.

Talk to your doctor or a registered dietitian before applying any of this if you:

- Have diabetes of any type
- Have coeliac disease, irritable bowel syndrome, or another digestive condition
- Have high blood pressure or kidney disease
- Are pregnant or trying to become pregnant
- Take regular medication
- Have a history of disordered eating
- Are under 18

None of what follows is written for those situations without professional supervision.

From the coach

You train seriously. You're up before the sun, you finish the intervals even when they burn, you put in the hours on the bike. You're building an engine.

The problem is that most athletes I coach arrive with the engine well built and the fuel improvised. They rehearse pace. They rehearse cadence. They rehearse transitions. Nutrition gets decided the night before.

This guide exists to close that gap. It won't tell you how to eat in daily life. It will tell you **how much, when, and in what form to put fuel into and around your sessions**, with numbers you can use tomorrow.

One rule organises everything else: **the gut is trainable, and nothing is new on race day.**

The athlete who rehearses fuelling with the same discipline as intervals stops surviving races and starts racing them.

Let's get to work.

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1. Why fuel decides the session

Your body burns fat and carbohydrate in proportions that shift with **one variable: intensity**.

At an easy pace, fat covers most of the work. You carry effectively unlimited fat stores, even if you're lean. But fat produces energy slowly. As you raise the intensity — a climb, an interval, race pace — your body shifts progressively toward carbohydrate, because carbohydrate produces energy fast.

The catch is that carbohydrate is a small tank. Between muscle and liver you store roughly 2,000 kcal of glycogen. On a long or hard session, that runs out.

What that means in practice, and it's all the physiology you need:

Situation	Consequence
Easy and short	Your stores cover it. You don't need to eat.
Hard or long	You spend glycogen faster than you can replace it.
Glycogen runs out	You slow down. That isn't weakness; it's biochemistry.

When the wall shows up at 20 miles, it didn't start there. It was built in the first 90 minutes, when you didn't eat enough.

The instruction: you don't fuel the session you're doing. You fuel the one you'll be doing in an hour.

2. Before the session

The rule: one gram per kilo, per hour remaining

It's the only formula worth memorising.

Carbohydrate (g) = your body mass (kg) × hours until the start

A 70 kg athlete:

Time before the start	Carbohydrate	Example
3–4 hours	210–280 g	White rice with chicken and a piece of fruit; a large bowl of oats with banana and honey
2 hours	140 g	Turkey sandwich on white bread + banana; bowl of cereal with low-fat milk
1 hour	70 g	Two slices of white bread with jam; a sports drink
30 minutes	35 g	A gel with water; half a ripe banana; a fruit pouch

The closer to the start, the less fibre and the less fat. Both slow gastric emptying, and whatever is still sitting in your stomach when the effort begins is what ruins the session.

The window worth avoiding

If you're taking carbohydrate before a hard effort, take it **more than 60 minutes out or in the final 10–15 minutes**. The middle ground — 30 to 45 minutes before — is where some athletes feel an energy dip in the first few kilometres.

It doesn't happen to everyone, and a proper warm-up blunts it. But if you've ever started a race feeling empty despite a good breakfast, that's the most likely reason, and the fix is free: move the gel.

Training fasted

It has a use, and it's a narrow one: **easy, short aerobic work**. There it's a legitimate tool.

For anything with intensity, training unfuelled will cap your pace and won't give you an adaptation you couldn't get while fed. This isn't a closed question in the literature — there are serious "train-low" protocols in use at the elite level — but for an athlete with a job, a family and 8 to 12 hours a week, the cost outweighs the return.

Simple rule: fasted on easy runs only. Everything else gets fuelled.

3. During the session

This is where the long session is won or lost. And it's where most athletes go short.

The ladder: how much per hour

Duration	Target per hour	Note
Under 45 min	Nothing	Water if it's hot.
45–75 min at high intensity	Mouth rinse, or 15–30 g	The benefit is central, not muscular. Swilling a sports drink for 5–10 seconds and spitting it out works.
1–2 hours	30–60 g	
2–3 hours	60–90 g	You'll need mixed sources. See below.
Over 3 hours	90 g, up to 120 g on the bike	Only with a trained gut.

The ceiling changes by discipline

Almost no guide separates this, and it's the difference between a plan that works and one that leaves you throwing up at 30 km.

Discipline	Practical ceiling	Why
Bike	100–120 g/h	No impact, you're seated, the stomach tolerates more.
Run	60 g/h baseline, up to 90 g/h with a trained gut	Impact and posture cut tolerance.
Swim	Not applicable	You can't eat while swimming. It gets solved before you get in.

Start early. At 20–30 minutes, not when you feel hungry. By the time you're hungry in a long session you're already behind, and clawing it back on a compromised stomach rarely works.

Above 60 g/h you need two sugars

Your gut absorbs glucose through one transporter and fructose through another. The glucose route saturates near 60 g/h. Push past that with a single sugar and the excess sits in your gut and causes trouble.

Look for products combining glucose (or maltodextrin) and fructose at a ratio near 1:0.8.

It's on the label of any modern product. With a single sugar, your real ceiling is 60 g/h no matter how much you train.

What to use

Context	Options
Running, or high intensity	Gels, chews, sports drink. Immediate digestion.
Bike, steady intensity	All of the above, plus bars, banana, boiled potato with salt, pressed rice cakes.
Last 20 min before running	Liquid only. Empty the stomach before you get off the bike.

Count in grams, not in units. A gel isn't a measurement: they range from 20 g to 40 g depending on brand. Read the label once, write the number down, and build your plan in grams. "Three gels an hour" means nothing; "75 grams an hour" means something.

4. Train your gut

This is the most important section in the guide.

You can know exactly how many grams you need and still be unable to absorb them, because your gut's capacity to process carbohydrate during exercise **is trainable, just like the engine**. An athlete who has never exceeded 40 g/h in training won't tolerate 90 g/h on race day. They'll tolerate nausea.

One honest caveat: the evidence on the exact mechanism is thinner than it's usually presented, and the studies aren't entirely consistent. What is consistent — and anyone who coaches long-distance athletes will confirm it — is that **repeated practice reduces symptoms and raises the amount you tolerate**. That's enough to build a protocol on.

The 6-week block

Run it **only on the week's long or race-specific session**. Never on a recovery run: there it does nothing useful and leaves you heavy for no return.

Weeks	Target per hour	How
1–2	30–40 g/h	One product only. The one you'll race with.
3–4	60 g/h	Add the second source (glucose + fructose).
5–6	80–90 g/h running · 100–120 g/h on the bike	At race intensity, with race products.

The four rules

1. **One variable at a time.** If you raise the amount, don't change the brand that week.
2. **Score symptoms 1 to 10** after every long session. Bloating, nausea, cramping, urgency.
3. **Above 5, you hold that dose another week.** You don't advance. The block stretches; it doesn't get forced.
4. **At race intensity, not easy pace.** Tolerating 90 g/h on a jog predicts nothing about tolerating it at marathon pace. The blood doing the digesting is the blood moving your legs.

When to back off

Vomiting, cramping that forces you to stop, or diarrhoea mid-session: **drop one step and stay there two weeks**. That isn't the plan failing, it's information. Today's ceiling is one rung lower than the one you tested.

And the rule that contains all of it

Nothing new on race day. Not a brand, not a flavour, not a concentration, not a breakfast time. If you're racing an Ironman on one brand of gel, you train on it for months first.

5. After the session

What's urgent isn't the session you finished. It's the next one.

The question that decides everything: when do you train again?

If you train again within 8 hours (a double day, or an early morning after an evening session), refuelling is a race against the clock:

- **Carbohydrate: 1.0–1.2 g/kg per hour for the first 4 hours.** For 70 kg that's 70–84 g an hour, across meals and drinks.
- **Protein: around 0.3 g/kg** in the first meal. For 70 kg, about 21 g.
- Start within the first 30–60 minutes.

If your next session is more than 8 hours away, the urgency disappears. A full meal in the next 1–2 hours does the job. Rice or potato with protein and vegetables. You don't need a shake.

On protein

Protein repairs; it isn't fuel. During exercise it supplies maybe 5–10% of energy, and that isn't its role.

What matters is **how much and how often, not the ratio to carbohydrate**. The old "3:1 carbs to protein" was an indirect way of reaching an adequate carbohydrate number. Aim at the number directly.

- **Per meal: 0.3–0.4 g/kg.** For 70 kg, 21–28 g.
- **Spread across the day**, not loaded into dinner.
- For reference, an endurance athlete in a training block sits around **1.8–2.0 g/kg per day**, with no weight goal attached.

Rehydration

Weighing yourself before and after tells you what you lost. Replace **125% to 150% of that loss** over the following hours, with sodium — not in one go, and not with plain water alone.

A kilo lost is 1.25–1.5 litres replaced, spread out. Drinking it all at once doesn't rehydrate you faster; it makes you urinate more.

6. Hydration and sodium

Drink to thirst, and know your number

These two don't contradict each other, and confusing them is the most common mistake here.

During the session: thirst is a good guide. It's the mechanism you have and it works. Forcing litres down because a schedule says so is precisely how you reach the problem in the next subsection.

Outside the session: measure your sweat rate, so you know whether thirst is leaving you short in extreme heat or very long efforts. It's a data point, not an order to drink.

The test, done properly

On a 60-minute session at training intensity:

1. **Weigh yourself naked before**, with an empty bladder.
2. **Train for an hour.** Record exactly how much fluid you drink, in ml.
3. **Weigh yourself naked after**, having towelled off the sweat.
4. **If you urinated during the session**, record it and subtract it.

Weight lost (g) + fluid drunk (ml) – urine (ml) = your sweat rate per hour. One gram of body mass equals one millilitre of fluid.

Repeat the test in different conditions. A number measured at 15 °C tells you nothing about a race at 30 °C. You need two or three, not one.

Sodium

Sodium is the electrolyte you lose in quantity through sweat, and its job is to help you hold on to the fluid you drink.

As a starting point: around 300 mg of sodium per hour on long sessions. More in heat — 500 to 700 mg/h or above. Less in cool conditions.

Two honest caveats:

- **Individual variation is enormous.** Some athletes lose three times as much sodium as others in the same session. If you finish with white salt marks on your kit and your cap, you're at the high end. If you never see salt, you're at the low end.
- **Salt capsules are not in this guide.** Dosing them depends on your sweat and on your cardiovascular and kidney health, and that isn't mine to decide. If you use them, do it with a sweat test or with your doctor — and rehearse them in training like everything else.

Cramps: what's known and what isn't

Cramps are commonly blamed on sodium. **The truth is that the question is open.** The competing explanation — neuromuscular fatigue, a muscle pushed past its capacity losing

control of its own contraction — has as much support as the sodium one, and both probably contribute depending on the case.

What can be said without overstating:

- **Magnesium during a race almost never resolves an acute cramp.** It's the most-taken supplement for this and the one with the least evidence at that particular moment.
- **The prevention that works is boring:** train at the intensity you intend to race, don't go out above your sustainable pace, and take sodium with your fluid on long, hot efforts.

Hyponatraemia: the one thing here that can be serious

Exercise-associated hyponatraemia happens when **you drink more than you lose** and dilute the sodium in your blood. It isn't dehydration — it's the opposite, and it's more dangerous.

The main cause is drinking too much. It isn't "forgetting your salt capsules" — taking sodium does not license unlimited drinking.

Warning signs during or after a long effort:

- A headache that keeps building
- Nausea or vomiting with no digestive cause
- Confusion, disorientation, speech that sounds off
- Swollen hands, feet or face; a ring that suddenly feels tight
- **Having gained weight during the race**

The rule: never finish a race heavier than you started. If that happens, you drank too much.

If these symptoms appear: stop drinking immediately and get medical help. This is not something to push through. And if you have a long race in heat ahead of you, with crowds and aid stations every mile — the classic setup — it's worth knowing this list before you start.

Smaller athletes are at higher risk, because the same volume of fluid dilutes more. If you weigh 50 kg, a 90 kg athlete's hydration plan is not your plan.

7. Carbohydrate loading

For races **over 90 minutes**. A 10k doesn't need it.

A plate of pasta the night before is not a load. A load is a 36 to 48 hour protocol.

The protocol

From three days out: start cutting fibre. Less raw vegetable, fewer legumes, less wholegrain, less fruit with skin. You don't want intestinal residue on race morning.

Final 24–36 hours: up to 10 g/kg per day of white, low-fibre carbohydrate, with lean protein.

For 70 kg that's up to 700 g of carbohydrate a day. It's far more food than you imagine, and it's why almost everyone believes they loaded and didn't.

- White rice, white pasta, white bread, peeled potato
- Honey, jam, juice, sports drinks
- Chicken, eggs, white fish
- Low fat, low fibre, nothing new

And ease off the training. Keep training hard and you burn what you're loading.

First time? Start at 5 g/kg

This matters and almost nobody says it. **A full load you've never rehearsed feels awful:** heaviness, bloating, a bad night's sleep before the race.

Start at **5 g/kg for your first race** and build across subsequent ones. The load gets trained like the gut does.

You'll gain weight, and that's fine

One or two kilos over those days is normal. Every gram of glycogen is stored with roughly three grams of water — the exact figure is debated, but that's the order of magnitude. **It isn't fat.** It's fuel plus its water, and that's water you're going to use.

8. Caffeine

It works. It's one of the few aids with solid evidence behind it, and it reduces perceived effort more than it changes what your muscle can do.

How I use it with my athletes

- **200 mg before the start.** Blood levels peak around 60 minutes.
- **Another 200 mg later** in long races — at T1 in a triathlon, or in the second half of a marathon, when mental fatigue arrives.
- That leaves roughly **300 mg circulating** when it counts.

For reference: an espresso is 60–80 mg, a caffeinated gel 25–50 mg, a can of cola about 35 mg.

The part almost nobody does: building up to it

Start at 100 mg in training and build gradually, race by race. The final amount depends on your body mass and your tolerance, and the only way to find it is in sessions, not on a start line.

This is exactly what we do with training load: progression, not a jump.

Before you use it

Don't use caffeine as a race strategy if you:

- Have arrhythmias, uncontrolled high blood pressure, or any cardiac condition — speak to your doctor first
- Are pregnant
- Have an anxiety disorder that caffeine aggravates
- Are under 18

And for everyone:

- **Watch the accidental stacking.** Coffee plus caffeinated gels plus cola at the aid stations adds up fast, and overshooting doesn't improve performance — it adds a racing heart, gut trouble and a sleepless night.
- **If you drink coffee daily, don't cut it abruptly** in race week. A withdrawal headache on race morning is a self-inflicted problem.
- **Never use it for the first time on race day.**
- For evening races, remember it will affect your sleep that night.

About the products you're using

Gels, powders, drinks and capsules are supplements, and supplements carry contamination risk. **If you compete under a federation with anti-doping testing, use only third-party certified products** (Informed Sport, NSF Certified for Sport). Caffeine isn't banned, but it is on WADA's monitoring programme.

And the general rule behind all of it: **food first, product second**. A gel solves a logistics problem, not a diet one.

9. Heat and altitude

Heat

Heat changes three things at once:

1. **You sweat more** — your sweat rate can double. Measure in heat, not only in cool weather.
2. **You need more sodium** — 500 to 700 mg/h or above.
3. **Your stomach empties more slowly**, because blood is diverted to the skin to shed heat.

The practical consequence is counter-intuitive: in heat, dilute. The same carbohydrate in more water. If your drink is concentrated and it's 32 °C, the stomach shuts down. And if you're planning to push past the middle of the per-hour range, heat is the condition where you're most likely not to tolerate it.

Lower your grams-per-hour target when it's very hot. Absorbing 60 g beats failing to absorb 90 g.

Altitude

Above about 2,000 m:

- **You burn more carbohydrate** at the same relative intensity. Your pace drops and the glycogen cost doesn't.
- **You lose more fluid** — dry air, higher breathing rate — and often without feeling thirsty.
- Add carbohydrate and add fluid relative to your sea-level plan.

Iron at altitude is a medical question, not a section in this guide. Altitude exposure raises iron demands, and supplementing without bloodwork isn't a coach's decision. If you train or live at altitude, raise it with your doctor.

10. When your stomach shuts down

It will happen. The question isn't whether, it's what you do.

Diagnosis, after the session

If you finish with diarrhoea or gut discomfort, there are two likely causes and they're easy to tell apart:

- 1. Product tolerance.** You usually already know: one brand or flavour always disagrees with you. Swap it and move on.
- 2. Concentration versus intake rate.** The common one, and the one nobody calculates. **Recompute the grams against the minutes you actually consumed them in**, not against the full hour. If you took 90 g in 20 minutes, your gut didn't receive 90 g/h — it received the equivalent of 270 g/h during that window. The fix is to dilute more, or to spread it better.

Mid-race

When the stomach shuts down, in this order:

- 1. Slow down.** It's the only thing that returns blood flow to the gut. Non-negotiable, and first.
- 2. Drop the solids.** Move to liquid.
- 3. Dilute.** More water for the same carbohydrate.
- 4. Cut the caffeine** for the rest of the race.
- 5. Walk the aid stations.** Swallowing while running makes everything worse.
- 6. Lower your hourly target** and accept less. Finishing on 40 g/h beats abandoning on 90.

Once it settles, build back gradually. Don't try to recover what you missed: it doesn't come back, and the attempt is what causes the second episode.

And what I tell my athletes: will it happen in the race? Yes. That's why we train it.

11. Race templates

How to use these: they're starting points to rehearse, not race-day law. Practise each one at least twice in specific sessions before you use it in competition.

The numbers are in **grams of carbohydrate**, not in units of product. To convert, read the label on what you use: a typical gel is 20 to 40 g, a bottle of sports drink 30 to 60 g depending on concentration.

5k and 10k

Duration: 20 to 75 minutes. Focus: start full, touch nothing.

When	What
Dinner before	Normal, carbohydrate-based, low fibre. No excess.
Breakfast, 2–3 h out	2–3 g/kg. Toast with jam, oats, banana.
45 min out	Sips of water or sports drink. Don't load fluid.
During	Nothing. Water at aid stations if it's hot.

If the effort runs past 45 minutes and it's all-out, a mouth rinse with sports drink in the second half has an effect. You don't have to swallow it; the benefit is neural.

Half marathon

Duration: 1:15 to 3:00. Focus: stable blood glucose for the closing push.

When	What
Dinner before	Carbohydrate increased, low fibre. Normal salt with the meal.
Breakfast, 3 h out	3 g/kg. Oats with banana, or a bagel with honey.
Final 10–15 min	20–25 g with water, if you've tested it.
During	60–90 g/h from minute 20. In practice: 20–25 g every 20 minutes.
Rough total	120–200 g depending on your time
Hydration	To thirst. Sports drink at most stations, water at the rest.

Marathon

Duration: 2:45 to 6:00. Focus: energy management. Loading is mandatory.

When	What
48 h out	Load: up to 10 g/kg/day (5 g/kg if it's your first). Fibre down from three days out.
Breakfast, 3 h out	3 g/kg, rehearsed in training.
Final 10–15 min	20–25 g with water.
During	60–90 g/h from minute 20. 20–25 g every 20 minutes, none skipped.
Sodium	~300 mg/h. More in heat.
Caffeine	200 mg before the start; the rest from 25–30 km.
Hydration	To thirst, alternating sports drink and water.

The most common marathon mistake isn't eating too little at the end. It's eating too little in the first hour, when you feel perfect and it doesn't seem necessary.

Time trial / 40k

Duration: ~1 hour, at threshold. Focus: liquid energy, zero logistics.

- **One bottle with 30–60 g** of sports drink. Small sips every 10–15 minutes.
- Solids: none. Chewing pulls you out of position and you don't need it.
- A gel taped to the frame just in case. Ideally you don't use it.

Gran fondo / 100k+

Duration: 3 to 6+ hours. Focus: the stomach tolerates it — use that.

Segment	What
Hours 1–2	60–90 g/h , up to 120 g/h if you've trained the gut. Solids: bars, jam sandwich, banana, boiled potato with salt.
Hours 3+	Same target, shift to easy digestion: gels, chews, drink.
Hydration	One bottle of sports drink + one of water per hour. Adjust for heat.
Sodium	~300 mg/h, more in heat.

Sprint triathlon

Duration: 1:00 to 1:45. Focus: speed. Don't lose time eating.

- **Breakfast, 2 h out:** 2 g/kg, easy to digest.
- **Swim:** nothing. A sip of water before you get in.
- **T1:** a mouthful of sports drink to clear the salt or chlorine taste.
- **Bike (20k):** one bottle of sports drink. Nothing else.

- **T2:** if your 5k will take more than 30 minutes, 20–25 g here *and only if you've tested it*. Any faster and it won't take effect in time.
- **Run (5k):** nothing. Water if it's hot.

Olympic triathlon

Duration: 2:00 to 3:30. Focus: the bike feeds the run.

- **Breakfast, 2–3 h out:** 2–3 g/kg.
- **Swim:** 20–25 g fifteen minutes before the start if it's been more than 3 h since breakfast.
- **Bike (40k):** this is where you eat. **One bottle of sports drink + 40–60 g in gels**, split: one at 20 minutes, another before T2.
- **T2:** a mouthful of water.
- **Run (10k):** carry 20–25 g and take it around 5 km if energy dips. Water to thirst.

Ironman 70.3

Duration: 4:00 to 7:00. Focus: the bike is the window. Don't eat there and you won't run the half.

Segment	What
Breakfast, 3 h out	3 g/kg, rehearsed, low fibre.
Swim	20–25 g fifteen minutes before the start.
T1	A mouthful of sports drink. No rush.
Bike (90k)	60–90 g/h. First 15 min liquid only. Hour 1: solids. Hours 2–3: gels, chews, drink. Rough total: 180–300 g.
Final 15 min of the bike	Stop the solids. Empty the stomach before you run.
Run (21k)	45–60 g/h. 20–25 g every 25–30 minutes.
Aid stations	Walk them to drink properly. Alternate water and sports drink.
Sodium	~300 mg/h; 500–700 mg/h in heat.

Ironman

Duration: 9 to 16 hours. Focus: crisis management, patience, and flavour fatigue.

Segment	What
Breakfast, 3–4 h out	3–4 g/kg, high carbohydrate, low fat.
Swim	20–25 g before the start.
Bike km 0–90	80–90 g/h while the gut is fresh. Mix sweet and savoury: pretzels, boiled potato, a small sandwich. This is your best window of the day.
Bike km 90–180	60–70 g/h. Lower the target as tolerance falls, and shift to gels and drink. Your digestive system tires; don't ask more of it exactly when it can give less.
Special needs	Something you genuinely like, to break the flavour fatigue.
Run (42k)	45–60 g/h. Gels, broth for the salt, cola late on.
If the stomach shuts down	Section 10. Slow down first.
Cola	The final 10–15 km: sugar and caffeine in a format that almost always goes down. Flat if you tolerate it better that way.

On "use what the race offers": broth and cola work, but **they're new products if you've never tried them.** Test them on a long ride first. The nothing-new rule doesn't make an exception for plan B.

12. The female athlete

You're not a smaller man, and your plan isn't built by dividing his.

Fuel in every phase

The first decision, and the one that pays most: **meet your carbohydrate targets in every phase of your cycle.** A great deal has been written about metabolic differences between men and women, and the real evidence is weaker and more contradictory than it's usually presented. What isn't in dispute is that going short on carbohydrate hurts anyone.

The numbers in sections 2, 3 and 5 are your numbers. Don't adjust them down for sex.

Hydration

Lower sweat rates and smaller bodies mean **higher hyponatraemia risk**, because the same volume of fluid dilutes more.

Drink to thirst. Use sodium on long sessions. Don't copy the hydration plan of a 90 kg training partner, and re-read section 6.

Iron

Female endurance athletes have a clearly higher risk of iron deficiency, and deficiency wrecks performance because it reduces oxygen transport. It's the most common cause of "I'm training the same and getting worse."

What I can tell you: it's a real risk, it's common, and it's worth ruling out before anything else when performance drops without explanation.

What I can't tell you: whether you're deficient, which tests to ask for, or what to take. **Don't supplement iron without bloodwork and a medical recommendation.** Excess iron is toxic, and taking it blindly masks the real cause of fatigue for months.

Action: if performance drops without explanation, see your doctor and raise iron. That's where it gets decided.

When your body signals an energy shortfall

This is where my job ends and another professional's begins, and I'd rather say so directly.

When a body trains a lot and takes in little energy over a sustained period, it starts shutting functions down to save. It happens to men and women, but in women one signal appears earlier and more clearly.

Signs worth paying attention to:

- **Your menstrual cycle becomes irregular, changes, or stops**
- You get ill often: colds that linger, recurring infections
- Stress fractures, or repeated bone injuries
- Fatigue that doesn't lift with rest days
- Performance declining month on month despite training the same or more
- Feeling cold constantly, sleeping badly, low mood

If you recognise yourself in this, it isn't something we adjust in the training plan. It's an appointment with your doctor and a registered dietitian, and the sooner the better: the bone and hormonal consequences are far easier to prevent than to reverse.

I can adjust your loads and support you through it. The rest isn't my ground, and a coach who tells you otherwise is doing you no favours.

13. What to do now

Don't try to apply all of this next week. In order:

1. **This week:** calculate your pre-session breakfast with the formula in section 2. One gram per kilo per hour until the start. It's the cheapest change and the most noticeable.
2. **This week:** run the sweat-rate test in section 6, once.
3. **The next 6 weeks:** run the gut block in section 4. That's the section that changes your races.
4. **Your next long race:** use the template for your distance, exactly as you rehearsed it.
5. **Read section 6 in full before your next long race in heat.** The hyponatraemia symptoms are the one thing in this guide worth knowing by heart.

If you want this fitted to you

This guide gives you the system. What it can't give you is your number: your sweat rate, your gut tolerance, your segment-by-segment plan for your actual race, in your climate, at your pace.

That's what **1-to-1 Endurance Coaching** is for: the swim, bike and run plan, the specific strength work, and a fuelling strategy grounded in your reality and your event.

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Let's get to work.

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Founder & Head Coach, Triaperformance

Sources

The numbers in this guide come from the Triaperformance coaching methodology and from the following references:

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